

easyWave: fast tsunami simulation tool for early warning

Andrey Babeyko (GFZ German Research Centre for Geosciences, Potsdam). Contact: babeyko@gfz-potsdam.de

Short user starting guide

Stand: Aug 2013

.....

Practical recommendations to organize simulations

easyWave is a 'unix-style' command-line driven tool. Unlike its previous version, current version does not require configuration file – model runtime and output parameters have been assigned to their default values which, however, can be changed at a command-line.

Actually, you need only one single executable called *easyWave* (or *easyWave.exe* for Windows) to run simulations. However, to help with the visualization of results, I provide another tool called *ew2png*. This tool also has to be run with command-line arguments, and there are many of them. To make things easier, I provide two shell-scripts requiring minimal user interaction to visualize (i) maximum wave heights- *sshmax2png.sh* (or *sshmax2png.cmd* for Windows) and (ii) snapshots of wave propagation- *ssh2png.sh* (or *ssh2png.cmd*). Important! The two shell-scripts assume that the *ew2png* converter is located in *~/easyWave* (or *C:\easyWave* for Windows).

Keeping this in mind, I recommend to organize your simulations in the following way.

On Linux machine create *~/easyWave* directory in your home, download and unpack *easyWave.xxx.linux64.tar.gz* into this directory.

On Windows machine create *C:\easyWave* directory, download and unpack *easyWave.xxx.windows.zip* into this directory.

Create sub-directory */grids* for sample bathymetry grids and download them from the corresponding ftp-location.

Create sub-directory */sources* for sample initial conditions and download them from the corresponding ftp-location.

Create sub-directory */pois* for sample Point-Of-Interest files and download them from the corresponding ftp-location.

Create sub-directory */run* for your simulation(s).

Starting simulation

If you start the *easyWave* executable without any parameters, you'll see help screen describing program usage and all possible command-line options.

Basically, you need to provide a minimal input to start a simulation. And that is: (1) bathymetry grid; (2) initial conditions; (3) computation time. All other parameters are optional.

For example, if you follow my recommendations, copy some bathymetry grid from *easyWave/grids* into *easyWave/run* (e.g. *e2r4Indian.grd* – Indian Ocean etopo2v2 re-sampled to 4 arc minutes). Then copy sample initial conditions file from *easyWave/sources* into *easyWave/run* (e.g. *BengkuluSept2007.flt* – source model for

the 12 Sept 2007 M8.4 Bengkulu earthquake by Lorito et al. 2008 presented as a set of Okada's faults). Then open a terminal window (on Windows: command-prompt by Start->Run->cmd) and go to the `~/easyWave/run` directory (cd `C:\easyWave\run`).

Since now *easyWave* executable is located one level above, start it with:

```
../easyWave -grid e2r4Indian.grd -source BengkuluSept2007.flt -time 120
```

In few seconds you should get output consisting of:

eWave.2D.XXXXXX.ssh – wave height snapshot at time XXXXXX (in seconds)

eWave.2D.sshmax – map of maximal wave heights

eWave.2D.time – map of arrival times

easywave.log – log file

(If something goes wrong, you may also get an *error.msg* file with error messages)

Have a look into the *easywave.log* – it lists all used runtime parameters.

Visualization of results

Use the *sshmax2png.sh* shell-script to create a PNG-image with maximum wave heights. Just type:

```
../sshmax2png.sh
```

without any parameters for default scaling. This script starts the *ew2png* binary converter with predefined color palette and scale. Inspect this shell-script if you want to use another palette or change other parameters. Run *ew2png* without arguments to see its help screen.

Default scaling for maximum wave heights is from 0.1 to 10 meters (logarithmic scale). You can change it by command line arguments. E.g.:

```
../sshmax2png.sh 0.01 1
```

Use the *ssh2png.sh* shell-script to create PNG-images with snapshots of wave propagation. This script requires at least one command-line parameter which is a time label of a snapshot (*eWave.2D.XXXXXX.ssh* files). Retrieve this time label from the filename you would like to visualize. E.g.:

```
../ssh2png.sh 03600
```

will create a PNG-file called *eWave.2D.03600.png* based on the snapshot output *eWave.2D.03600.ssh* corresponding to the time stamp of 1 hour (=3600 seconds).

The command:

```
../ssh2png.sh all
```

will process all available time-frames and create a MPEG4-animation of wave propagation (Linux only!). It is assumed that you have “avconv” utility on your PATH.

Default scaling for these snapshots is +/-0.5 meter (also logarithmic scale). You can change it from command line. E.g.:

```
../ssh2png.sh 03600 5
```

to scale wave heights color table to 5 meters.

Computing mareograms and impact summary at selected positions

In addition to 2D-output, easyWave can produce also output at selected positions – at Points-Of-Interest (poi). Sample POIs files are located in *easyWave/pois* sub-directory. Copy *poiIndonesia.poi* file to your run directory and start *easyWave* with following command line:

```
../easyWave -grid e2r4Indian.grd -source BengkuluSept2007.flt -poi poiIndonesia.poi -time 120
```

After running the model, you should find two new poi-output files:

eWave.poi.ssh – mareograms (meters) as ASCII table with the first column presenting time in minutes

eWave.poi.summary – impact summary with estimated time of arrival (minutes) and estimated coastal maximum wave height (meters)

Mareograms and coastal maximum wave heights undergo Green's law amplification from POI-closest grid-node to 1 meter depth. To suppress shoaling effect add '0' after poi-coordinates in poi-input file (useful for deep ocean buoys).

Remark: You can also get POI-loading statistics if using *-poi_report* option at a command line.

Visualization of mareograms

Use whatever you want – *eWave.poi.ssh* is just an ASCII-table.

Visualization of tsunami impact on POIs (Points Of Interest)

Use *poi2png*. See *poi2png.sh* as an example.

easyWave command line syntax and parameters

Start *easyWave* without command-line arguments to get built-in help.

Remarks on input file format

- bathymetry (*.grd) and 2D-output (*.2D.ssh, *.2D.sshmax, *.2D.time)
Golden Software (C) text or binary GRD format

- initial conditions

Either:

file with (multiple-)Okada fault parameters,

or:

map of sea surface elevation in Golden Software (C) GRD format.

Specifying Okada's fault.

There is no fixed set of fault parameters, but you should give enough parameters to set up a fault.

Required parameters:

-mw value or **-slip** value (slip in meters)

-location lon lat depth (lon and lat in decimal degrees, depth in positive kilometers)

-strike value

-dip value

-rake value

Optional parameters:

-refpos string ('C'- fault center (default) | 'MT'- mid of top-edge | 'BT'- begin of top-edge | 'MB'- mid of bottom edge | 'BB'- begin of bottom-edge)

-size length width (in km). Note: size is optional if **-mw** is given. In this case default size would be estimated from the JMA scaling laws. If you did specify **-slip**, you must explicitly specify length and width.

-rigidity value (in Pa, default = 3.5e10 Pa)

Examples:

-mw 8.0 **-location** 100 10 30 **-rake** 90 **-dip** 10 **-strike** 330

-slip 10 **-size** 100 50 **-location** 100 10 30 **-rake** 90 **-dip** 10 **-strike** 330 **-rigidity** 2.e10

- POIs (Points-of-interest, forecast points)

Text file with each row containing at least 3 values: (1) POI-ID, (2) longitude, (3) latitude.

Optional fourth parameter takes value of 1 (default, in this case may be omitted) or 0 and tells the program to extrapolate wave height from POI location onto the nearby coast using coastal amplification law (usually, Green's law).

Usually, we would like to set “0” for buoys or other deep ocean POIs.

Example:

Padang 100.30 -0.966

Bengkulu 102.200 -3.766

DART23401 88.54 8.91 0

